



PraFFL: A Preference-Aware Scheme in Fair Federated Learning

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Abstract

Fairness in federated learning has emerged as a critical concern, aiming to develop an unbiased model among groups (e.g., male or female) of diverse sensitive features. However, there is a trade-off between model performance and fairness, i.e., improving model fairness will decrease model performance. Existing approaches have characterized such a trade-off by introducing hyperparameters to quantify client's preferences for model fairness and model performance. Nevertheless, these approaches are limited to scenarios where each client has only a single pre-defined preference, and fail to work in practical systems where each client generally has multiple preferences. To this end, we propose a **Preference-aware scheme in Fair Federated Learning** (called PraFFL) to generate preference-specific models in real time. PraFFL can adaptively adjust the model based on each client's preferences to meet their needs. We theoretically prove that PraFFL can offer the optimal model tailored to an arbitrary preference of each client, and show its linear convergence. Experimental results show that our proposed PraFFL outperforms six fair federated learning algorithms in terms of the model's capability of adapting to clients' different preferences. Our implementation is available at <https://github.com/rG223/PraFFL>.

CCS Concepts

• **Computing methodologies** → **Distributed artificial intelligence**.

Keywords

Fairness, Federated Learning, Trade-Off, Preference-Aware, Model Adaptation, Theoretical Guarantee

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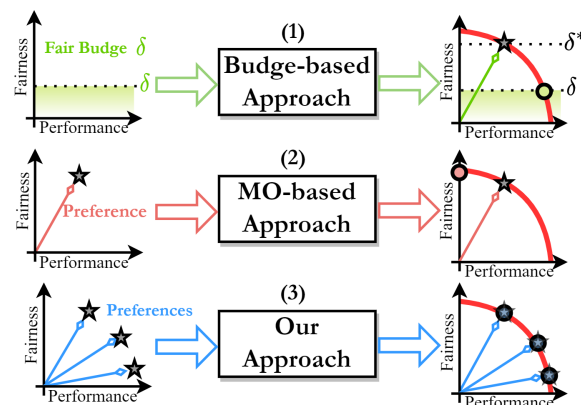


Figure 1: The model trained under different approaches. Gray stars represent the desired models, and colored circles represent the obtained models. The red curve indicates the optimal performance-fairness trade-offs, typically unknown in advance. δ^* is the fair budget that most closely aligns with the green preference arrow. For the first case, the desired model is obtained when $\delta = \delta^*$, where δ^* is the fair budget that most closely aligns with the green preference arrow. However, since the Pareto front is unknown, it is not possible to determine the exact value of δ^* that corresponds to the intersection of the preference direction and the Pareto front. Instead, δ must be repeatedly adjusted until it reaches δ^* . For the second case, the obtained model deviates from the desired model due to the concave shape of the Pareto front (see Fig. 3). For the third case, our proposed approach not only achieves the desired models but also enables simultaneous handling of multiple preferences.

1 Introduction

Federated learning (FL) is a distributed machine learning method that learns from data across multiple clients [27, 38, 40–42], where a global model is trained cooperatively among clients without sharing data. Since the global model learns from multiple data sources, it has a stronger generalization ability across multiple clients than separately training model for each client [44]. However, a key concern in FL is to ensure that the trained model is unbiased towards any particular groups (e.g., special races) of sensitive features [17, 21, 33]. This implies that a fair model needs to treat different groups equally, such that model's predictions are identical across different groups.

Fair FL was proposed to improve the fairness of the model [1, 11, 32, 46], where the fairness of the model is usually defined as the differences in predictions across different groups. Specifically, the smaller the prediction differences among different groups, the

Table 1: Time complexity comparison of multiple FFL methods with respect to the number of preferences κ .

Method	Time Complexity	Real-Time Adaption	Preference Awareness
LFT+Ensemble [46]	$O(\kappa)$	✗	✗
LFT+Fedavg [22]	$O(\kappa)$	✗	✗
Agnosticfair [9]	$O(\kappa)$	✗	✗
FairFed [11]	$O(\kappa)$	✗	✗
FedFB [46]	$O(\kappa)$	✗	✗
PraFFL (Ours)	$O(1)$	✓	✓

better the model’s fairness. Unfortunately, many studies [9, 15, 43] argue that improving model fairness and model performance in FL is a trade-off issue, i.e., model fairness and model performance cannot be improved simultaneously. Some existing works characterized the trade-off using hyperparameters. For example, budget-based approaches (e.g., LFT+Fedavg [47] and Fairfed [11]) treat the fairness of the model as a constraint (see Fig. 1). Multi-objective-based (MO-based) approaches (e.g., FAIR-FATE [36]) takes the linear combination of model performance and model fairness as the optimization objective. However, most of aforementioned works have two main limitations. First, their determined models may not satisfy a client’s given preference over model fairness and model performance. Specifically, those budget-based approaches aim to match fairness but ignore performance. For example, Fig. 1 depicts that the budget-based approaches need to repeatedly train the model after adjusting fair budget δ until the desired model is obtained. Those MO-based approaches (e.g., FAIR-FATE) can obtain the desired model only when the shape of the optimal performance-fairness curve (also called Pareto front) is convex [3], while such a convexity is not necessarily satisfied in FL scenario. Second, these approaches (e.g., [1, 11, 36]) may not be applicable to the scenario where each client has multiple preferences. A practical example is to take the social platform TikTok as a client to meet preferences from billions of users [18]. In particular, users expect the model to recommend high-quality videos that align with their interests, i.e., these users prioritize model performance. In contrast, some other users seek unbiased recommendations towards/against any particular racial group, i.e., these users prioritize model fairness. For those aforementioned works (e.g., [1, 11, 32]), when each client has different requirements (i.e. preferences), their model needs to be retrained multiple times to adapt each requirement. Thus, their time complexity grows linearly with the number of preferences, as shown in Table 1. Inspired by this, we aim to answer the following research questions:

How can the model (i) effectively align with each client’s specific preference and (ii) be adaptively adjusted in real time based on each client’s diverse preferences?

These problems involve three technical challenges: (I) establishing the connection between the two objectives (performance and fairness) and the client’s preference; (II) alleviating the impact of data heterogeneity; and (III) preventing the privacy leakage of client preferences. To this end, we propose a **Preference-Aware** scheme in Fair Federated Learning (PraFFL). Fig. 1(c) shows that our proposed PraFFL can accurately learn a mapping from the preference

vector to a trade-off point between model performance and model fairness, enabling the model to link its fairness and performance with the corresponding preference vector. In PraFFL, we include a personalized FL method to alleviate the impact of data heterogeneity. Meanwhile, we introduce a hypernetwork that isolates each client’s preference information, preventing it from being exchanged with other clients or the server, thereby ensuring client privacy. PraFFL can provide the preference-specific model to each client and achieve the time complexity of $O(1)$. In other words, once the model is trained by PraFFL, it allows the model to adapt to each client’s preferences during the inference phase. In addition, PraFFL is theoretically proven to obtain the Pareto front within one training.

The main contributions of this paper are highlighted as follows:

- We present a preference-aware learning framework, which enables the model to adapt to different preferences of each client.
- We propose a PraFFL scheme to further address the three technical challenges while allowing the model to adapt to arbitrary preferences of each client.
- We theoretically prove that PraFFL can linearly converge to optimal model that meets each client’s preference, and can learn the Pareto front on each client’s dataset.
- Numerical experiments validate that our proposed PraFFL outperforms other baselines on four datasets in terms of the model’s capability in adapting to clients’ different preferences.

2 Related Work

There are two types of fairness in fair FL: client-based fairness [24, 25, 30, 39] and group-based fairness [8, 30, 45]. The purpose of client-based fairness is to reduce the variance of model performance across all clients while maintaining the performance of these models. Our work mainly discusses group-based fairness, in which each client has some sensitive features (e.g., gender and race). The goal of group-based fairness is to avoid discrimination against sensitive features [20, 35, 48]. Recent studies [9, 11, 34] investigated how to achieve group fairness in FL. For example, an adaptive FairBatch debiasing algorithm [35] was proposed under the FL framework, where the optimization problem of each client is a bi-level optimization problem of model fairness and sampling probability for different groups. FedVal [28] is another global reweighting strategy, where the client’s aggregation weight depends on the fairness budget of the model in the validation set. However, previous studies have shown that improving model performance and model fairness is a trade-off problem [11, 34, 36, 46]. To control the trade-off between model performance and fairness, the most common approach is to treat fairness as a constraint in optimizing each client’s model [9, 14, 49]. Specifically, FAIR-FATE [36] introduces a weight coefficient to linearly combine fairness and model performance as the final optimization goal. Furthermore, Fairfed [11] uses the FairBatch debiasing algorithm [35] in the optimization stage to ensure fairness, and a reweighting method in the aggregation stage to control the performance-fairness trade-off. Most of the above methods can only obtain a model corresponding to one of the client’s preferences. It is challenging for these methods to meet multiple user’s preferences.

3 Problem Formulation

3.1 Multi-Objective Learning

Let $\mathbf{b} \in \mathcal{B}$ and $y \in \mathcal{Y}$ denote the non-sensitive features and labels of dataset $\mathcal{D}$, respectively. $a \in \mathcal{A}$ represents the sensitive feature where a supports the inclusion of multi-dimensional sensitive information, for example, young man for both age and gender dimensions. The feature vector can be further expressed as $\mathbf{x} \triangleq (a, \mathbf{b}) \in \mathcal{X}$. According to [4, 46], we consider a binary classification problem (i.e., $y \in \mathcal{Y} \triangleq \{0, 1\}$, where $y = 1$ indicates positive class) and multi-group sensitive feature (i.e., $a \in \mathcal{A} \triangleq \{0, \dots, m\}$). Now, we introduce two objectives in fair FL, i.e., model performance and model fairness.

Model Performance. Let $\theta : \mathcal{X} \rightarrow \mathcal{Y}$ be a classifier. Usually, the cross-entropy loss function is used to quantify the model performance in the classification problem, and it is defined as follow:

$$\mathcal{L}_1(\mathbf{x}, y, \theta) = -[y \log(\theta(\mathbf{x})) + (1 - y) \log(1 - \theta(\mathbf{x}))], \quad (1)$$

where $\theta(\mathbf{x})$ represents probability of predicting a data sampled with feature $\mathbf{x}$ belongs to the positive class.

Model Fairness. A widely adopted criterion for assessing classifier fairness is *DP disparity* [12]. Let $\hat{y}$ represent the the label predicted by the classifier based on feature $\mathbf{x}$. DP disparity quantifies the maximum difference in expectation of the classifier θ that predicts a data sample to be in positive class across different sensitive groups

$$\max_{j \in \mathcal{A}} |\mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}_k} [\hat{y} = 1 | a = j] - \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}_k} [\hat{y} = 1]|. \quad (2)$$

Eq. (2) is not suitable for optimization (in gradient-based training) due to its discontinuity. Instead, model fairness is usually characterized by the following loss function [46]

$$\mathcal{L}_2(\mathbf{x}, y, \theta) = [(a - \bar{a})(\theta(\mathbf{x}) - \bar{\theta}(\mathbf{x}))]^2, \quad (3)$$

where $\bar{a}$ and $\bar{\theta}(\mathbf{x})$ represent the average values of a and $\theta(\mathbf{x})$ over $\mathcal{D}$, respectively. $\mathcal{L}_2$ quantifies the correlation between sensitive features and the model predictions. A higher value of $\mathcal{L}_2$ suggests that sensitive features are more strongly correlated with the model's predicted probabilities, indicating a lower fairness.

Then, we convert the fair FL into a multi-objective problem to solve, which can be expressed as follows:

$$\min_{\theta \in \Theta} \mathcal{L}(\mathbf{x}, y, \theta) = (\mathcal{L}_1(\mathbf{x}, y, \theta), \mathcal{L}_2(\mathbf{x}, y, \theta)), \quad (4)$$

where $\mathcal{L}(\mathbf{x}, y, \theta) : \mathcal{X} \times \mathcal{Y} \times \Theta \rightarrow \mathbb{R}_{\geq 0}^2$ is a two-dimensional objective vector. For presentation simplicity, in the rest of this paper, we use the terms “solution” and “trained model” interchangeably.

Model performance and model fairness are conflicting, and we need to make trade-offs between these two objectives [34, 46]. For ease of notation, let us denote j -th objective as $O_j(\theta) = \mathcal{L}_j(\mathbf{x}, y, \theta)$ for all $j \in \{1, 2\}$. We give the following definitions of fair FL in the context of multi-objective optimization [5].

DEFINITION 3.1 (PARETO DOMINANCE). Let $\theta^a, \theta^b \in \Theta$, θ^a is said to dominate θ^b , denoted as $\theta^a < \theta^b$, if and only if $O_i(\theta^a) \leq O_i(\theta^b)$ for all $i \in \{1, 2\}$ and $O_j(\theta^a) < O_j(\theta^b)$ for some $j \in \{1, 2\}$. In addition, θ^a is said to strictly dominate θ^b , denoted as $\theta^a <_{\text{strict}} \theta^b$, if and only if $O_i(\theta^a) < O_i(\theta^b)$ for all $i \in \{1, 2\}$.

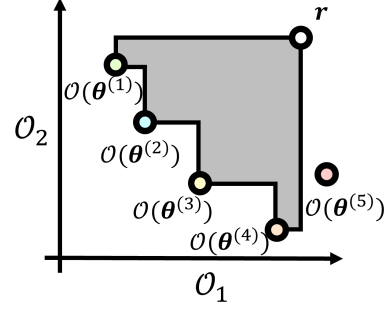


Figure 2: Diagram of hypervolume.

DEFINITION 3.2 (PARETO OPTIMALITY). $\theta^* \in \Theta$ is Pareto optimal if there does not exist $\hat{\theta} \in \Theta$ such that $\hat{\theta} < \theta^*$. In addition, θ' is weakly Pareto optimal if there is no $\theta \in \Theta$ such that $\theta <_{\text{strict}} \theta'$.

DEFINITION 3.3 (PARETO SET/FRONT). The set of all Pareto optimal solutions $\mathcal{P}^* \subseteq \Theta$ is called the Pareto set, and its image in the objective space $\mathcal{PF}^* = \{O(\theta) \mid \theta \in \mathcal{P}^*\}$ is called the Pareto front.

To evaluate the quality of a solution set, a hypervolume (HV) indicator [13] is commonly used.

DEFINITION 3.4 (HYPERVOLUME). Let $\hat{\Theta} = \{\theta^{(1)}, \dots, \theta^{(n)}\}$ be a set of objective vectors given solution $\theta^{(i)}$, $i \in [n]$. The hypervolume indicator $\mathcal{H}_r(\hat{\Theta})$ is the two-dimensional Lebesgue measure of the region dominated by $\mathcal{P}^*(\hat{\Theta})$ and bounded by a pre-defined reference point $\mathbf{r} \in \mathbb{R}^2$.

Fig. 2 shows the calculation of HV for a set of five solutions, where the gray area represents the HV value. We aim to make the model adapt to client preferences, and to maximize the HV of the Pareto front in the objective space.

3.2 Preference-Aware Learning Framework

Inspired by the scalarization method in multi-objective optimization, we introduce preference vector $\lambda \in \Lambda \in \mathbb{R}_{>0}^2$ ($\sum_{i=1}^2 \lambda_i = 1$) and incorporate it into the original multi-objective optimization problem (Eq. (4)) to characterize clients' preference information. In particular, we consider a generalized setting [23] where each client k aims to obtain its own model θ_k . This setting can generalize the conventional FL case where clients aim to learn an identical global model. To enable the model to adapt to each client's individual preference, we concatenate ($\oplus$) the input $\mathbf{x}$ with the preference vector λ as joint features for training each client's local model $\theta_k : \mathcal{X} \oplus \Lambda \rightarrow \mathcal{Y}$. By considering the preference vector in FL system, Eq. (4) can be transformed to

$$\min_{(\theta_1, \dots, \theta_K) \in \Theta_K} \frac{1}{K} \sum_{k \in [K]} \mathbb{E}_{\lambda \sim \Lambda} \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}_k} \lambda \cdot \mathcal{L}(\mathbf{x} \oplus \lambda, y, \theta_k), \quad (5)$$

where Θ_K is the feasible space of K models. $\mathcal{D}_k$ represents the dataset of client k . The system aims to provide models tailored to any preference. Hence, the system must be trained on the expectation of preference distribution Λ to encompass all potential preferences. In practice, the distribution Λ can be set to a uniform distribution when there is no prior knowledge of clients. Once the model θ_k ($k \in [K]$) is optimized, the prediction results of each

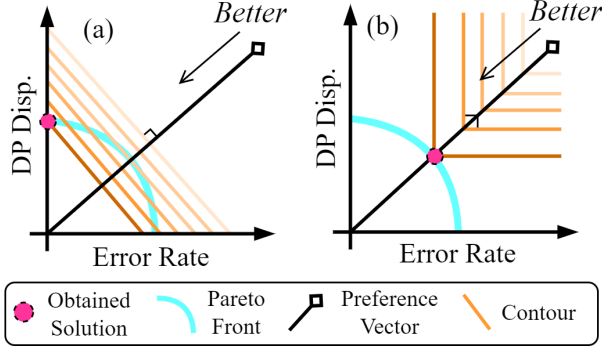


Figure 3: The solution obtained by different optimization functions. (a): The weighted-sum function (Eq. (5)). (b): The weighted Tchebycheff function (Eq. (7)).

client's dataset will be determined by each client's given preference in the inference phase. The Pareto front can be generated by examining the models of all possible preference vectors in the inference phase (i.e., one preference generates one Pareto optimal solution).

Although our proposed preference-aware learning framework (Eq. (5)) takes client preferences into account, there are three difficulties in obtaining high-quality Pareto front by Eq. (5):

- (I) We have no prior knowledge of the Pareto front shape. According to [3], a weight-sum function in Eq. (5) only works when the Pareto front is convex. Eq. (5) cannot guarantee the Pareto optimal solution is in the direction of preference vector when the Pareto front is concave (illustrated in Fig. 3(a)).
- (II) The data distribution among different clients is heterogeneous, that is, their Pareto fronts are also different (depending on each client's data distribution).
- (III) Optimizing the system via Eq. (5) may leak preference information of clients (e.g., data reconstruction attack [37]) because preference vectors are used as input data.

4 The Proposed Method: PraFFL

In this section, we propose a preference-aware scheme in fair FL (PraFFL) to further address three difficulties in solving Eq. (5). Section 4.1 presents a preference alignment approach to address difficulty (I). Moreover, Section 4.2 incorporates client preferences into personalized FL to address difficulty (II). Furthermore, we propose to use a hypernetwork to consider preferences in Section 4.3, which protects the clients' preference privacy (addressing difficulty (III)). In addition, Section 4.4 summarizes the algorithm details of PraFFL scheme.

4.1 Alignment of Model with Preference

It is difficult to provide a model whose performance and fairness fall in the direction of the client's corresponding preference vector. To obtain a Pareto optimal solution in the direction of any given preference vector, we introduce a weighted Tchebycheff function [29] as the optimization function for each client as follows:

$$\min_{\theta_k} g_t(\mathbf{x} \oplus \boldsymbol{\lambda}, y, \theta_k | \boldsymbol{\lambda}) = \min_{\theta_k} \max_{i \in \{1,2\}} \left\{ \frac{\mathcal{L}_i(\mathbf{x} \oplus \boldsymbol{\lambda}, y, \theta_k)}{\lambda_i} \right\}. \quad (6)$$

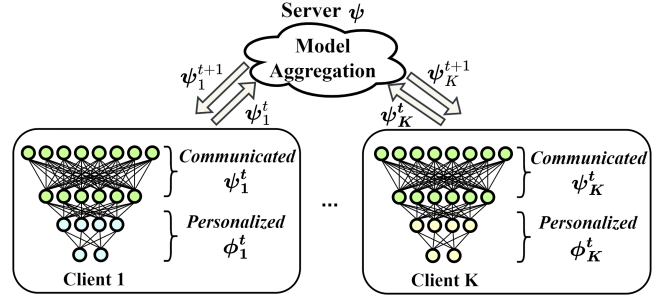


Figure 4: Personalized federated learning framework.

We can obtain the following promising property of problem (6).

LEMMA 4.1 (PREFERENCE ALIGNMENT [29]). *Given a preference vector $\boldsymbol{\lambda} > 0$, a solution $\theta_k \in \Theta$ is weakly Pareto optimal to problem 4 if and only if θ_k is an optimal solution to problem (6).*

Lemma 4.1 shows that given a preference vector $\boldsymbol{\lambda}$, our method can obtain a weakly Pareto optimal solution for each client if problem (6) is minimal. Then, substituting Eq. (6) into problem (5), the optimization problem in our system can be represented as

$$\min_{(\theta_1, \dots, \theta_K) \in \Theta_K} \frac{1}{K} \sum_{k \in [K]} \mathbb{E}_{\boldsymbol{\lambda} \sim \Lambda} \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}_k} g_t(\mathbf{x} \oplus \boldsymbol{\lambda}, y, \theta_k | \boldsymbol{\lambda}). \quad (7)$$

Eq. (7) can optimize each model towards the direction of the preference vector. Fig. 3(a) shows the result obtained by weight-sum function (Eq. (5)), which cannot accurately obtain the model on the preference vector. In Fig. 3(b), the weight Tchebycheff optimization function we introduced can accurately obtain the model on the preference vector if Eq. (7) is minimum. In this way, we can align the model to client's given preference vector. However, due to the data heterogeneity of clients, problem (7) cannot reach the minimum value, resulting in the failure to meet the condition of Lemma 4.1. In the following subsections, we introduce personalized FL based on hypernetwork mapping, which allows each client to obtain its own optimal model without revealing the privacy of client preferences.

4.2 Personalized FL Based on Preferences

To address difficulty (II) in Section 3, we present a personalized FL framework (see Fig. 4). In this framework, each client's model is divided into two parts (communicated model and personalized model). The communicated model $\psi : \mathbb{R}^{|\mathcal{X}|} \rightarrow \mathbb{R}^d$ is the part of the model that is shared between clients. d is the dimension of the data after passing through the communicated model. Each client also has a personalized model $\phi_k : \mathbb{R}^d \rightarrow \mathbb{R}^{|\mathcal{Y}|}$. The model of client k is expressed as $\theta_k = (\psi, \phi_k)$. The inclusion of the personalized model can solve difficulty (II) because each client's personalized model is optimized according to its corresponding data distribution. Thus, by taking into account both the communicated and personalized models, the optimization goal of PraFFL can be transformed into

$$\min_{(\phi_1, \dots, \phi_K) \in \Phi_K} \frac{1}{K} \sum_{k \in [K]} \mathbb{E}_{\boldsymbol{\lambda} \sim \Lambda} \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}_k} g_t(\mathbf{x} \oplus \boldsymbol{\lambda}, y, \theta_k | \boldsymbol{\lambda}), \quad (8a)$$

$$\text{s.t. } \psi = \arg \min_{\psi \in \Psi} \frac{1}{K} \sum_{k \in [K]} \mathbb{E}_{\boldsymbol{\lambda} \sim \Lambda} \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}_k} \mathcal{L}_1(\mathbf{x} \oplus \boldsymbol{\lambda}, y, \theta_k). \quad (8b)$$

Our idea is to ensure the performance of the communicated model (Eq. (8b)), and then optimize the personalized model according to the given preferences (Eq. (8a)). As will be seen in later sections, we consider an iterative optimization process. When Eq. (8b) is being optimized, clients collaboratively optimize a global communicated model ψ to minimize cross-entropy loss (performance) over their own datasets. After Eq. (8b) has been optimized, the system optimizes the personalized models of clients, considering the expected model performance and fairness over clients' preferences. Note that the optimization of communicated model takes into account the preference vector as input. These preference vectors are easily leaked through data reconstruction (difficulty (III)) [19].

4.3 Coupled Preferences with Hypernetwork

To further solve difficulty (III), we introduce a *hypernetwork* to prevent the leakage of the client's preference information. Hypernetwork is deployed on every client. Let $\beta_k : \mathbb{R}^m \rightarrow \mathbb{R}^{|\phi_k|}$ be a hypernetwork, which aims to learn a mapping from the preference vector to the parameters of the personalized model. Given a preference vector λ , the personalized model ϕ_k of client k is determined as follows:

$$\phi_k = \beta_k(\lambda). \quad (9)$$

Based on Eq. (9), client's preference λ are represented as the personalized model via a hypernetwork rather than as part of the joint features (as mentioned in Eq. (5)). Each client k can directly obtain its personalized model by specifying vector λ in Eq. (9). As shown in Fig. 5, if the hypernetwork can be well-trained, then it can generate the personalized model that satisfies the corresponding preference in the inference phase. The model of client k is further expressed as $\theta_k(\lambda) = (\psi, \beta_k(\lambda))$. The optimization problem of our system is transformed into the following

$$\min_{(\beta_1, \dots, \beta_K) \in \mathcal{B}_K} \frac{1}{K} \sum_{k \in [K]} \mathbb{E}_{\lambda \sim \Lambda} \mathbb{E}_{(x,y) \sim \mathcal{D}_k} g_t(x, y, \theta_k(\lambda) | \lambda), \quad (10a)$$

$$s.t. \psi = \arg \min_{\psi \in \Psi} \frac{1}{K} \sum_{k \in [K]} \mathbb{E}_{\lambda \sim \Lambda} \mathbb{E}_{(x,y) \sim \mathcal{D}_k} \mathcal{L}_1(x, y, \theta_k(\lambda)). \quad (10b)$$

Eq. (10a) aims to learn a mapping relation from preference vector distribution Λ to a set of personalized models. Due to the client's preferences being learned in the hypernetwork, each client's preference information is isolated from other clients. Therefore, hypernetwork can learn each client's preferences while protecting the client's preference information (addressing difficulty (III)). So far, the three difficulties mentioned in Section 3 have been solved.

4.4 PraFFL Algorithm

Now, we propose PraFFL algorithm to solve problem (10), which can generate a preference-specific model for each client while addressing difficulties (I)–(III). Specifically, there are T rounds in total. In each round, a proportion of $p \in (0, 1]$ clients is selected for training, where the set of clients is denoted as $\mathcal{S}_t$. Then, the communicated model and the hypernetwork are updated sequentially at each client. Each client k first downloads the global communicated model ψ as its local communicated model ψ_k .

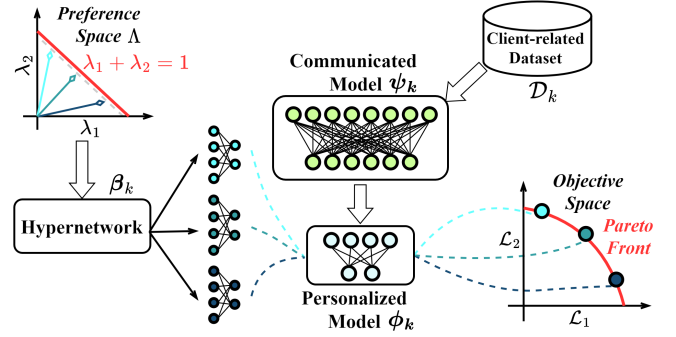


Figure 5: The illustration of hypernetwork inference for client k . Hypernetwork β_k returns the corresponding personalized models to client k based on the preference vectors.

To stabilize the optimization of the communicated model for solving problem (10b), we need to freeze the personalized model when the communicated model is being updated. In particular, personalized model of client k is frozen to $\phi_k = \beta(\tilde{\lambda})$, where $\tilde{\lambda}$ is chosen to be $(\frac{1}{2}, \frac{1}{2})$ so that the personalized model is unbiased towards model performance and fairness when optimizing the local communication model. Client k performs τ_c times of gradient descent algorithm for updating the communicated model at t -th round

$$\psi_k^{t,s+1} = \psi_k^{t,s} - \eta \mathbb{E}_{(x,y) \sim \mathcal{D}_k} \nabla \psi_k \mathcal{L}_1(x, y, \theta_k^{t,s}(\tilde{\lambda})), \quad (11)$$

where $s = 1, \dots, \tau_c$ and η is the learning rate.

After updating the communicated model, each client then optimizes the hypernetwork. When optimizing problem (10a), client k freezes the communicated model as ψ_k^{t,τ_c+1} . The optimization of the hypernetwork for client k can be expressed as follows:

$$\min_{\beta_k} \mathbb{E}_{\lambda \sim \Lambda} \mathbb{E}_{(x,y) \sim \mathcal{D}_k} g_t(x, y, \theta_k(\lambda)). \quad (12)$$

Eq. (12) is difficult to optimize because there are infinite possible values for λ . We use the Monte Carlo sampling method in stochastic optimization to optimize Eq. (12). Thus, client k performs τ_p

$$\beta_k^{t,j+1} = \beta_k^{t,j} - \eta \mathbb{E}_{(x,y) \sim \mathcal{D}_k} \frac{1}{n_\lambda} \sum_{i=1}^{n_\lambda} \nabla \beta_k g_t(x, y, \theta_k^{t,j}(\lambda^{(i)})), \quad (13)$$

where $j = 1, \dots, \tau_p$, n_λ is the batch size for sampling preference vectors and $\lambda^{(i)}$ is sampled from preference vector distribution Λ . In our work, we set Λ to be a uniform distribution.

Once the communicated model and personalized model of participated clients are updated at t -th round, the participated clients send their communicated models ψ_k ($k \in [K]$) to the server. Afterwards, the server updates the communicated model with average aggregation as follows:

$$\psi_k^{t+1,1} = \frac{1}{|\mathcal{S}_t|} \sum_{k \in \mathcal{S}_t} \psi_k^{t,\tau_c+1}. \quad (14)$$

After the model aggregation in Eq. (14), the communicated models at $(t+1)$ -th round are the same across all clients. In contrast, the personalized model is customized based on the specific data distribution of each client. As shown in Fig. 5, each client can choose arbitrary preference vectors, and the hypernetwork then provides

the corresponding personalized models to the client. Then, each client can combine the personalized model with the communicated model, and the combined model can produce the desired results considering the client's preference over model performance and model fairness.

5 ALGORITHMIC ANALYSIS

In this section, we analyze performance guarantee of the trained model and the convergence rate of PraFFL. The proofs of this section are provided in the Appendix.

5.1 Learning Capabilities

Based on the properties of the weighted Tchebycheff function [29], we determine the following three main results:

- Existence of the Pareto optimal model;
- Condition of Pareto model being Pareto optimal;
- The ability to find the Pareto front.

We analyze the model of an arbitrary client k . Given a preference vector λ in problem (10a), i -th objective can be expressed as $\mathcal{L}_i(\mathbf{x}, y, \theta_k(\lambda))$, $\forall i \in \{1, 2\}$ over dataset $\mathcal{D}_k$. First, we analyze the existence of the Pareto optimal model.

LEMMA 5.1 (EXISTENCE). *Given a preference vector $\lambda \in \mathbb{R}_{\geq 0}^2$, problem (10) has at least one Pareto optimal solution.*

Lemma 5.1 shows that PraFFL has Pareto optimal solutions for any client's preference λ . After we determine the existence of Pareto optimality of the model, we then analyze the condition when a model is Pareto optimal.

LEMMA 5.2 (PARETO OPTIMALITY). *Given a preference vector $\lambda \in \mathbb{R}_{\geq 0}^2$, a model is weakly Pareto optimal if it minimizes problem (10). The model is Pareto optimal if and only if the optimal solution of problem (10) is unique.*

This Lemma shows that PraFFL can achieve the Pareto optimal solution for the preference vector λ if the conditions in Theorem 5.2 are satisfied. Finally, we analyze the ability of PraFFL to find the Pareto front, and we have the following theorem:

THEOREM 5.3 (PARETO FRONT). *Let $\theta'_k \in \Theta$ be an optimal model for a given client's preference. Then, there exists a preference vector $\lambda \in \mathbb{R}_{\geq 0}^2$ such that we can obtain $\theta_k^*(\lambda) = \theta'_k$ from problem (10).*

Theorem 5.3 shows that the models can obtain the entire Pareto front. This is because, for any Pareto optimal model θ'_k , there is a preference vector λ that the model $\theta_k^*(\lambda)$ is Pareto optimal.

5.2 Convergence Analysis

In this subsection, we discuss the convergence properties of PraFFL. The optimization between the communicated model and the personalized model are correlated. We first prove that the communicated model converges linearly under the personalized FL framework. We then prove that the convergence rate of the personalized model is the same as that of the communicated model. Finally, we conclude that PraFFL also converges linearly.

The convergence analysis of problem (10) is difficult because the weighted Tchebycheff function is non-differentiable. To overcome

this issue, we introduce a *smooth Tchebycheff function* as follows:

$$\min_{\theta_k} \tilde{g}(\mathbf{x}, y, \theta_k(\lambda) | \lambda) = \min_{\theta_k} \frac{1}{\gamma} \log \sum_{i \in [2]} e^{\frac{\gamma \mathcal{L}_i(\mathbf{x}, y, \theta_k(\lambda))}{\lambda_i}}, \quad (15)$$

where $\gamma > 0$ represents the smooth factor.

Before presenting convergence results, we list the following two commonly used assumptions.

ASSUMPTION 5.1. *The expectation of stochastic gradients is always bounded, i.e., for all $\phi_k^t \in \Phi$ and $\psi^t \in \Psi$, we have $\mathbb{E}[\|\nabla \tilde{g}\|^2] \leq G^2$.*

ASSUMPTION 5.2. *$\mathcal{L}_i(\mathbf{x}, y, \theta_k(\lambda))$ is μ_i -strongly convex and L_i -smooth for all $k \in [K]$, $\lambda \in \Lambda$, and $\theta_k(\lambda) \in \Theta$.*

Based on the above assumptions, we can deduce the strong convex coefficient and smoothness coefficient of the smooth Tchebycheff function.

LEMMA 5.4 (CONVEXITY OF SMOOTH TCHEBYCHEFF FUNCTION). *If Assumption 5.2 is satisfied for $i \in [1, 2]$, then $\tilde{g}(\mathbf{x}, y, \theta_k(\lambda))$ is $2\log(\sum_{i \in [2]} e^{\frac{\mu_i}{2\lambda_i}})$ -strongly convex function, for $k \in [K]$ and $\lambda \in \Lambda$.*

LEMMA 5.5 (SMOOTHNESS OF SMOOTH TCHEBYCHEFF FUNCTION). *If Assumption 5.2 is satisfied for $i \in [1, 2]$, then $\tilde{g}(\mathbf{x}, y, \theta_k(\lambda))$ is $\sum_{i \in [2]} \frac{L_i}{\lambda_i}$ -smooth, for $k \in [K]$.*

LEMMA 5.6 (CONVERGENCE OF THE COMMUNICATED MODEL [6]). *If the communicated model ψ is optimized by Fedavg [27] and given a constant $\zeta > 0$, then ψ converges to optimal communicated model ψ^* at a linear rate*

$$z(t) = \mathbb{E}[\|\psi^t - \psi^*\|^2] \leq (1 - \eta\zeta)^{t/2} \mathbb{E}[\|\psi^1 - \psi^*\|], \quad (16)$$

with a probability no smaller than $1 - te^{-100 \min(|X|^2 \log(|S_t|), d)}$.

Lemma 5.6 implies that in order to achieve a bound of $(1 - \eta\zeta)^{t/2} \mathbb{E}[\|\psi^1 - \psi^*\|] \leq \varepsilon$, we must run $O(\log(\frac{1}{\varepsilon}))$ iterations of gradient descent. This convergence rate is often called *linear convergence*.

For simplicity, let us denote $\check{\mu} = 2\log(\sum_{i \in [2]} e^{\frac{\mu_i}{2\lambda_i}})$ and $\check{L} = \sum_{i \in [2]} \frac{L_i}{\lambda_i}$. The convergence speed of the personalized model is given by the following proposition.

PROPOSITION 5.7 (PROGRESS OF ONE STEP). *Under Assumptions 5.1 and 5.2, let client k get selected with probability p_k at each communication round t with decaying $\eta_t = \frac{2\check{\mu}}{(t+1)p_k}$, we have*

$$\begin{aligned} & \mathbb{E}[\|\phi_k^{t+1} - \phi_k^*\|^2] \\ & \leq (1 - \frac{2}{t+1}) \mathbb{E}[\|\phi_k^t - \phi_k^*\|^2] + \frac{8\check{\mu}^2 G \check{L}}{(t+1)^2 p_k^2} \sqrt{\mathbb{E}[\|\psi^t - \psi^*\|^2]} \\ & \quad + \frac{4\check{\mu}^2 G^2}{(t+1)^2 p_k^2} + (\frac{4\check{\mu}^2 \check{L}^2}{(t+1)^2 p_k^2} + \frac{2\check{\mu}}{t+1} (\check{L} - \check{\mu})) \mathbb{E}[\|\psi^t - \psi^*\|^2] \\ & \quad + \frac{4\check{\mu} \check{L}}{t+1} \sqrt{\mathbb{E}[\|\phi_k^t - \phi_k^*\|^2] \mathbb{E}[\|\psi^t - \psi^*\|^2]}. \end{aligned} \quad (17)$$

Proposition 5.7 shows the convergence speed of the personalized model is affected by the convergence speed of the communicated model, which is intuitively reasonable because these two types of models are optimized sequentially in our system. We further prove that each client's model $\theta_k(\lambda)$ also is linear convergence.

Table 2: Local/Global HV results of seven different algorithms on four datasets. Local HV is evaluated on the local client datasets, and Global HV is evaluated on the global dataset. The best results are highlighted in bold, while the second-best results are underlined.

Method	Global HV $\uparrow$			
	SYNTHETIC	COMPAS	BANK	ADULT
LFT+Ensemble	0.479	0.555	0.881	0.764
LFT+Fedavg	<u>0.698</u>	0.539	<u>0.883</u>	0.768
Agnosticfair	0.530	0.546	0.882	<u>0.783</u>
FairFed	0.342	0.507	0.878	0.269
FedFB	0.599	0.517	<u>0.883</u>	0.764
EquiFL	0.642	<u>0.564</u>	0.877	0.761
PraFFL	0.763	0.629	0.889	0.828

Method	Local HV $\uparrow$			
	SYNTHETIC	COMPAS	BANK	ADULT
LFT+Ensemble	0.468	0.514	0.138	0.501
LFT+Fedavg	0.479	<u>0.555</u>	0.881	0.764
Agnosticfair	0.530	0.551	0.880	<u>0.784</u>
FairFed	0.342	0.507	0.878	0.269
FedFB	0.599	0.517	<u>0.883</u>	0.763
EquiFL	<u>0.604</u>	0.526	0.882	0.764
PraFFL	0.778	0.631	0.895	0.834

THEOREM 5.8 (CONVERGENCE OF PRAFFL). *If there exists a constant A such that $\frac{z(t+1)}{z(t)} \geq 1 - \frac{z(t)}{A}$, for $\lambda \in \Lambda$, then we have*

$$\mathbb{E}[\|\theta_k^t(\lambda) - \theta_k^*(\lambda)\|^2] = O(\log(\frac{1}{t})). \quad (18)$$

with a probability no smaller than $1 - te^{-100 \min(|\mathcal{X}|^2 \log(|S_t|), d)}$.

Theorem 5.8 shows that PraFFL linearly converges to Pareto optimal solution $\theta_k^*(\lambda)$.

6 NUMERICAL Experiments

In this section, we empirically compare our proposed PraFFL algorithm with six advanced baselines (LFT+Fedavg [22], LFT+Ensemble [46], Agnosticfair [9], FairFed [11], FedFB [46], and EquiFL [26]).

6.1 Experimental Settings

We report results from three runs on each dataset. PraFFL shows the solutions obtained by 1000 preference vectors in the experiment (in fact, any number of solutions can be obtained through PraFFL inference if a corresponding number of preference vectors are given). The total local epochs are set to 30 (i.e., $\tau_p + \tau_c = 30$). We use hypervolume (HV) [13] to measure the quality of the solution set obtained by each algorithm. We mainly report local HV on local datasets and global HV on global datasets in our experiments. The higher the quality of the solutions in satisfying the client's preferences, the larger the HV value. Four widely used datasets are employed to conduct experiments, including SYNTHETIC [46], COMPAS [2], BANK [31] and ADULT [10].

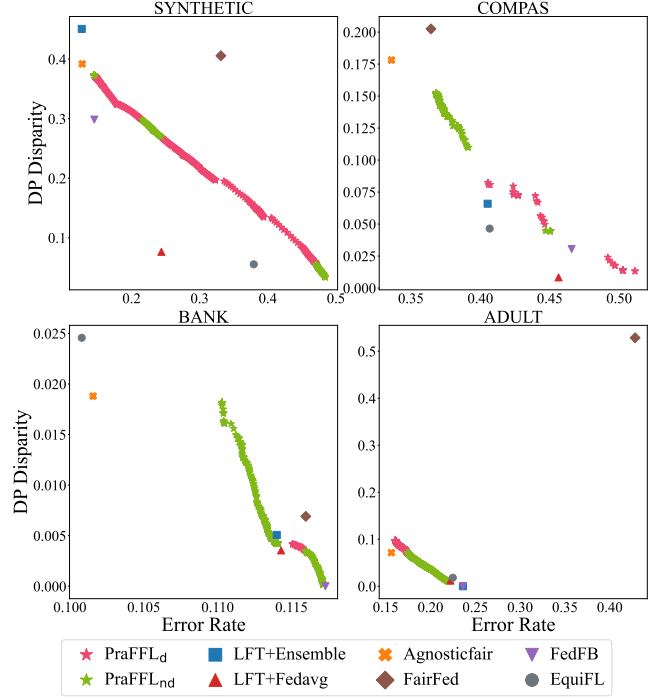


Figure 6: Comparison of solutions obtained by six algorithms on four datasets. We divide the solution set obtained by the PraFFL method into two categories: PraFFL_d and PraFFL_{nd} . PraFFL_d means that the solution is dominated (i.e., there is another algorithm that is better than the solutions in PraFFL_d), and PraFFL_{nd} means that the solution is non-dominated (i.e., there is no other solution that is better than the solutions in PraFFL_{nd}).

6.2 Main Results

Inspired by [7, 16], we evaluate all algorithms on local and global datasets. The results presented in Table 2 demonstrate that PraFFL is consistently superior to the performance of five baselines regarding HV. Our proposed PraFFL has the best local HV value on the four datasets, which are 0.778, 0.631, 0.895, and 0.834, respectively. This is because our proposed method can infer any number of personalized models for clients, thereby obtaining more solutions with different trade-offs between model performance and fairness. Additionally, Table 2 also shows the global HV of each algorithm. Although PraFFL is designed for local Pareto front, it can achieve the best global HV compared with six other baselines. Recall that other baselines require retraining the model if each client has more than one preference, and the training time increases linearly with the number of preferences (see Table 1).

Moreover, Fig. 6 shows the final solutions obtained by the baselines and the non-dominated solutions obtained by PraFFL among 1000 solutions. According to the dominance relationship in Definition 3.1, we divide the solutions obtained by PraFFL into two categories (PraFFL_d and PraFFL_{nd}). PraFFL_d (red stars) are inferior to some baselines in some preferences. The solution set in PraFFL_{nd} (green stars) is superior to other baselines (it is the best trade-off

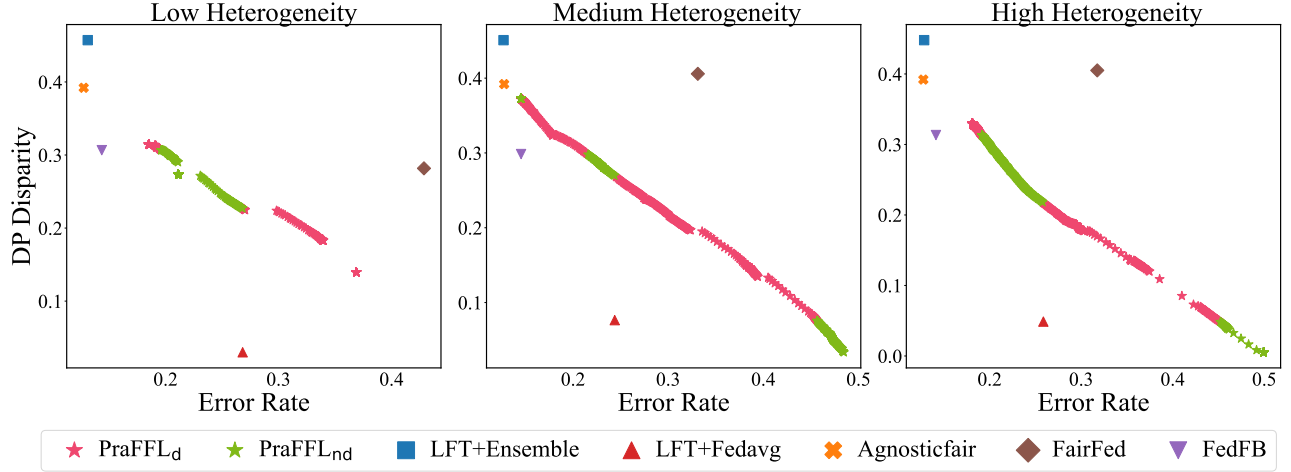


Figure 7: Accuracy-fairness solutions on the SYNTHETIC dataset with three clients having different heterogeneity distributions of the sensitive feature. Left: low heterogeneity with Dirichlet distribution $\text{Diri}(100)$. Middle: medium heterogeneity with Dirichlet distribution $\text{Diri}(5)$. Right: high heterogeneity with Dirichlet distribution $\text{Diri}(0.1)$.

between fairness and error rate for some specific preferences). Since the solution set of PraFFL covers the optimal trade-offs for most of the client's preferences, the solution set obtained by PraFFL is better than the solution obtained by other algorithms. The advantage of PraFFL is that it can provide arbitrary solutions for clients with only one training. In addition, the Pareto front obtained by PraFFL on the COMPAS and BANK datasets can be further improved. There are two aspects that can further improve the learned Pareto front: (I) The data distribution of the training and testing set are not exactly the same. Obtaining a superior solution set on the training set may not necessarily lead to a superior solution set on the testing set; (II) Our optimization goals of model performance and fairness are not directly equivalent to optimizing the error rate and DP disparity. Therefore, these two aspects lead to a gap between the solution set we obtained and the optimal solution set, which motivates future research directions.

6.3 Ablation Studies

This subsection mainly investigated the effects of data heterogeneity, number of clients, and hyperparameters in PraFFL.

6.3.1 The effect of the number of clients. Table 3 shows the results under different numbers of clients on the SYNTHETIC dataset. Two observations can be made: (I) Regardless of the large-scale number of clients, our proposed PraFFL achieves the best local HV and global HV compared to the five baseline algorithms, with PraFFL leading the best algorithm by 2% to 18%. This demonstrates that our PraFFL is highly effective in providing preference-specific models for clients in large-scale scenarios. (II) As the number of clients increases, there is a slight decrease in PraFFL's local HV and global HV. Nevertheless, PraFFL still achieves state-of-the-art performance compared with five baselines. This phenomenon could be attributed to the increasing divergence in local Pareto fronts with the growing number of clients, due to the differences in local datasets across

Table 3: Performance comparison of different methods across the number of clients on SYNTHETIC dataset.

Method	Local HV $\uparrow$			Global HV $\uparrow$		
	Number of Clients			Number of Clients		
	10	50	100	10	50	100
LFT+Ensemble	0.475	0.599	0.589	0.459	0.475	0.464
LFT+Fedavg	<u>0.663</u>	<u>0.614</u>	0.536	0.463	0.589	0.472
Agnosticfair	0.548	0.597	0.623	0.551	0.559	0.556
FairFed	0.415	0.394	0.388	0.372	0.309	0.329
FedFB	0.597	<u>0.614</u>	<u>0.649</u>	<u>0.613</u>	<u>0.591</u>	<u>0.589</u>
PraFFL	0.786	0.695	0.676	0.776	0.661	0.654

clients. As a result, this leads to the emergence of heterogeneous local Pareto fronts.

6.3.2 The effect of data heterogeneity. We examine the error rate and DP Disparity obtained by each algorithm under different heterogeneous levels on the sensitive features. As in [46], we study three heterogeneous scenarios. Fig. 7 shows the performance of the algorithms under three different heterogeneity levels. We can observe that our proposed PraFFL are robust than the other five baselines under different heterogeneous levels. Compared with the other five algorithms, the number of solutions of PraFFL_{nd} is larger, and such solutions are more widely distributed in different heterogeneities. For four baselines, the changes in the values of their two indicators under the three levels of heterogeneity are relatively small. It is worth noting that PraFFL is robust to large-scale user preferences. In other words, PraFFL not only have considerable performance in meeting users' dynamic preferences, but also maintain high stability under different degrees of data heterogeneity.

6.3.3 The effect of τ_p . HV is calculated by the solution set and a reference point (Fig. 2). HV is affected by two factors: (1) quality of the solution set (i.e., the closer to the origin, the higher HV);

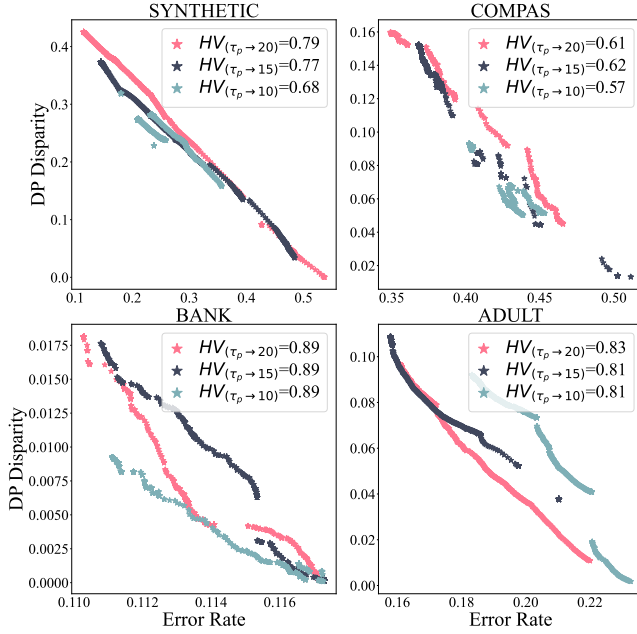


Figure 8: Solutions obtained by PraFFL under different τ_p .

and (2) diversity of the solution set (i.e., the broader the solutions distributed, the higher HV). Recall that $\tau_p + \tau_c$ equals the total local training epochs. As shown in Fig. 8, adjusting the value of τ_p is a trade-off problem. We can find that larger τ_p increases the diversity of the solution set, while it leads to a smaller τ_c , making the quality of the solution set worse. This is because the training of the communicated model mainly focuses on improving the solution set (Eq. (11)), and a larger τ_p will allow the personalized model to fully learn the user’s preferences. Conversely, a smaller τ_p decreases the diversity but improves the quality of the solution set. For instance, when $\tau_p = 10$ in COMPAS, the convergence results of the solution set are superior, but the diversity of the solution set is inferior.

6.3.4 Different preferences of clients. We analyze the mapping from different preference vectors to their corresponding solution set on four datasets. For the simplicity of verifying the effectiveness of PraFFL, we assume that the preferences of all users are unchanged during inference. In Fig. 9, we can find that the solutions corresponding to different preferences are clearly distinguishable. This shows that PraFFL can assign a unique solution to each user’s different preferences. In addition, we observe that the Pareto front obtained by PraFFL on SYNTHETIC, BANK, and ADULT have a wide range, which indicates that PraFFL provide models for various preferences of clients. The learned Pareto front under COMPAS is slightly worse than that under the other three datasets, which may be due to data quality. In general, PraFFL can meet most of the preference requirements of clients based on the preference vectors they provide.

7 Conclusion and Future Works

In this work, we proposed a preference-aware scheme in fair federated learning (PraFFL), which is capable of providing models

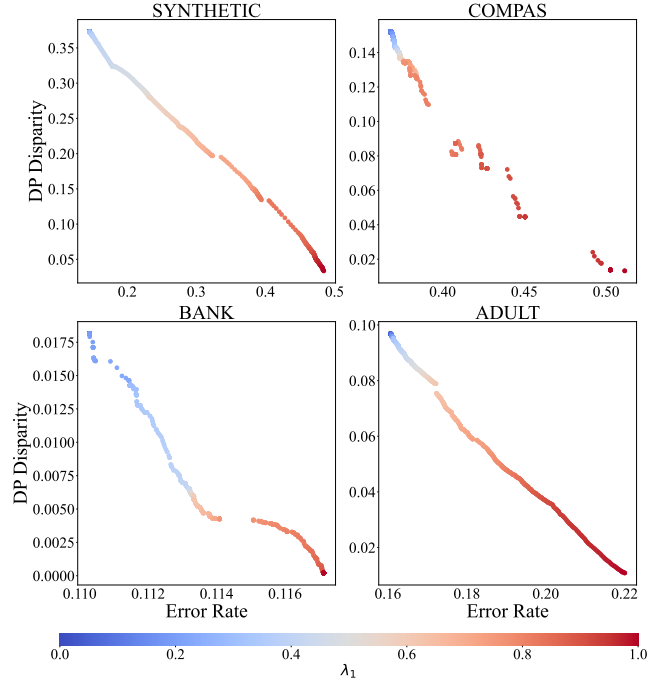


Figure 9: Mapping relationship from different preferences to model performance and fairness.

with different trade-offs to every client in real time. In the training phase, our proposed PraFFL does not require the assumption that the Pareto front is convex and preserves the client’s preference privacy. In the inference phase, the model obtained by PraFFL can be adaptively adjusted according to the client’s preferences. We theoretically prove that given a client’s preference, PraFFL can linearly converge to the Pareto optimal model and learn the entire Pareto front. Experimental results show that the solution set obtained by PraFFL on four datasets is well-performing, and the solutions obtained by PraFFL are the optimal choices for most of the client’s preferences compared to five advanced algorithms.

Although our proposed PraFFL can achieve promising results on the four datasets, it still has room for improvement. The first direction is to improve the generalization ability of the learned Pareto front. The distribution of the training and testing dataset is inconsistent. A good solution set on the training dataset may not lead to a superior solution set on the testing dataset. Therefore, the generalization of the learned Pareto front can be improved by regularization or using advanced data sampling strategies. In the second direction, the cross-entropy and fairness loss functions used in our work are not equivalent to the error rate and DP disparity formulations, respectively. To further improve the performance of PraFFL, a possible approach is to approximate the error rate and DP disparity formulations.

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A Appendix

A.1 Parameter Settings

The batch size of preference vectors n_λ is set to 64. The total communication rounds T is set to 10, and the batch size of data is 128 in the neural network training. We use Adam as optimizer and the learning rate η is 0.005. The smooth factor of smooth Tchebycheff function is set to 1. The reference point $\mathbf{r}$ in calculating hypervolume is set to (1, 1).

A.2 Datasets

According to the setting of [11, 46], the COMPAS and ADULT datasets have two clients, while the SYNTHETIC and BANK have three clients. In Section 6.3.1, we will conduct experiments on more clients. We split 30% of the data into the test set, and the other 70% of the data are split into the training set and validation set with a ratio of 9:1. Furthermore, we select the model that performs best in the validation set for testing.

A.3 Proofs

A.3.1 Proof of Lemma 5.4.

PROOF. Let $\tilde{g}(\phi_k)$ and $\mathcal{L}_i(\phi_k)$ be simply the parameter ϕ_k optimized by $\tilde{g}$ and $\mathcal{L}_i$, respectively. Let $z \in [0, 1]$, we start from the definition of strong convexity to obtain the strong convexity coefficient of $\tilde{g}$.

$$\tilde{g}_t(z\phi_k^{(a)} + (1-z)\phi_k^{(b)} \mid \lambda) \quad (19)$$

$$= \frac{1}{\gamma} \log \left(\sum_{i \in [3]} e^{\gamma(\mathcal{L}_i(z\phi_k^{(a)} + (1-z)\phi_k^{(b)}))/\lambda_i} \right) \quad (20)$$

$$\leq \frac{1}{\gamma} \log \left(\sum_{i \in [3]} e^{\gamma(z\mathcal{L}_i(\phi_k^{(a)}) + (1-z)\mathcal{L}_i(\phi_k^{(b)}))/\lambda_i} \right) \quad (21)$$

$$- \log \left(\sum_{i \in [3]} e^{\frac{\mu_i}{2\lambda_i}} \right) \cdot z(1-z) \|\phi_k^{(a)} - \phi_k^{(b)}\| \quad (22)$$

$$= \frac{z}{\gamma} \log \left(\sum_{i \in [3]} e^{\frac{\gamma\mathcal{L}_i(\phi_k^{(a)})}{\lambda_i}} \right) + \frac{1-z}{\gamma} \log \left(\sum_{i \in [3]} e^{\frac{\gamma\mathcal{L}_i(\phi_k^{(b)})}{\lambda_i}} \right) \quad (23)$$

$$- \log \left(\sum_{i \in [3]} e^{\frac{\mu_i}{2\lambda_i}} \right) \cdot z(1-z) \|\phi_k^{(a)} - \phi_k^{(b)}\|. \quad (24)$$

□

A.3.2 Proof of Lemma 5.5.

PROOF. Given loss function $\mathcal{L}_i(\mathbf{x}, y, \theta(\lambda))$ is L_i -smooth for $i = 1, 2$ and $\lambda \in \Lambda$, we now prove that the smooth Tchebycheff function $\tilde{g}(\mathbf{x}, y, \theta(\lambda) \mid \lambda)$ is $\frac{L_i}{\lambda_i}$ -smooth for both the personalized model ϕ_k and the communicated model ψ_k for $k = 1, \dots, K$. We take ϕ_k as an example, and let $\tilde{g}(\phi_k)$ and $\mathcal{L}_i(\phi_k)$ be simply the parameter ϕ_k optimized by $\tilde{g}$ and $\mathcal{L}_i$, respectively. We start from the definition of smoothness to obtain the smooth coefficient of $\tilde{g}$. Due to the chain rule $\nabla \tilde{g}(\phi_k) = \sum_{i \in [2]} \nabla \mathcal{L}_i(\phi_k) \nabla_{\mathcal{L}_i} \tilde{g}(\phi_k)$, we can calculate $\nabla_{\mathcal{L}_i} \tilde{g}(\phi_k) = \sum_{i \in [2]} \frac{e^{\mathcal{L}_i(\phi_k)}}{\lambda_i \sum_{i \in [2]} e^{\mathcal{L}_i(\phi_k)}}$. After that, we start from the

following equation:

$$\|\nabla \tilde{g}(\phi_k^{(a)}) - \nabla \tilde{g}(\phi_k^{(b)})\| = \left\| \sum_{i \in [2]} \nabla \mathcal{L}_i(\phi_k^{(a)}) \nabla_{\mathcal{L}_i} \tilde{g}(\phi_k^{(a)}) - \nabla \mathcal{L}_i(\phi_k^{(b)}) \nabla_{\mathcal{L}_i} \tilde{g}(\phi_k^{(b)}) \right\| \quad (25)$$

$$= \left\| \sum_{i \in [2]} \frac{e^{\mathcal{L}_i(\phi_k^{(a)})}}{\lambda_i \sum_{i \in [2]} e^{\mathcal{L}_i(\phi_k^{(a)})}} \nabla \mathcal{L}_i(\phi_k^{(a)}) - \sum_{i \in [2]} \frac{e^{\mathcal{L}_i(\phi_k^{(b)})}}{\lambda_i \sum_{i \in [2]} e^{\mathcal{L}_i(\phi_k^{(b)})}} \nabla \mathcal{L}_i(\phi_k^{(b)}) \right\| \quad (26)$$

$$\leq \sum_{i \in [2]} \frac{1}{\lambda_i} \|\nabla \mathcal{L}_i(\phi_k^{(a)}) - \nabla \mathcal{L}_i(\phi_k^{(b)})\| \quad (27)$$

$$\leq \sum_{i \in [2]} \frac{L_i}{\lambda_i} \|\phi_k^{(a)} - \phi_k^{(b)}\|^2. \quad (28)$$

The last step is due to the smoothness of $\mathcal{L}_i$. Then, we can also get the smoothness coefficient of $\tilde{g}(\phi)$,

$$\|\nabla \tilde{g}(\phi^{(a)}) - \nabla \tilde{g}(\phi^{(b)})\| \leq \sum_{i \in [2]} \frac{L_i}{\lambda_i} \|\phi^{(a)} - \phi^{(b)}\|^2. \quad (29)$$

□

A.3.3 Proof of Lemma 5.1.

PROOF. Suppose that none of the optimal solution (i.e., ϕ_k) in problem (10) is Pareto optimal. Let $\theta_k^*(\lambda) \in \Phi$ be an optimal solution to problem (10). Since we assume that it is not Pareto optimal, there must exist a model $\theta'_k(\lambda) \in \Phi$ which is not optimal for problem (10) but for which $\mathcal{L}_i(\mathbf{x}, y, \theta'_k(\lambda)) \leq \mathcal{L}_i(\mathbf{x}, y, \theta_k^*(\lambda))$, $\forall i \in \{1, 2\}$ and $\mathcal{L}_i(\mathbf{x}, y, \theta'_k(\lambda)) < \mathcal{L}_i(\mathbf{x}, y, \theta_k^*(\lambda))$, $\exists i \in \{1, 2\}$. Therefore, we have

$$\mathcal{L}_i(\mathbf{x}, y, \theta'_k(\lambda)) \leq \mathcal{L}_i(\mathbf{x}, y, \theta_k^*(\lambda)), \forall i \in \{1, 2\}. \quad (30)$$

We substitute this inequality into the smooth Tchebycheff function and get the following inequality

$$\frac{1}{\gamma} \log \sum_{i \in [2]} e^{\frac{\mathcal{L}_i(\mathbf{x}, y, \theta'_k(\lambda))}{\lambda_i}} < \frac{1}{\gamma} \log \sum_{i \in [2]} e^{\frac{\mathcal{L}_i(\mathbf{x}, y, \theta_k^*(\lambda))}{\lambda_i}}. \quad (31)$$

This means θ'_k is optimal of problem (10), but we suppose θ_k^* is optimal of problem (10). This contradiction completes the proof. □

A.3.4 Proof of Lemma 5.2.

PROOF. We suppose that $\theta_k^*(\lambda)$ is not Pareto optimal. By the definition of Pareto optimality (Definition 3.2), there exists another model $\theta'_k(\lambda) \in \Theta$ such that $\mathcal{L}(\mathbf{x}, y, \theta'_k) < \mathcal{L}(\mathbf{x}, y, \theta_k^*)$. Therefore, we have the following inequality

$$\mathcal{L}_i(\mathbf{x}, y, \theta'_k(\lambda)) \leq \mathcal{L}_i(\mathbf{x}, y, \theta_k^*(\lambda)), \forall i \in \{1, 2\}. \quad (32)$$

We also can obtain inequality (31), which contradicts the assumption, so $\theta_k^*(\lambda)$ is Pareto optimal for problem (10). □

A.3.5 Proof of Theorem 5.3.

PROOF. Given a preference vector $\lambda \triangleq (\lambda_i > 0, i \in \{1, 2\})$, let θ_k^* is defined as follows:

$$\theta_k^*(\lambda) \triangleq \arg \min_{\theta_k \in \Theta} \tilde{g}(x, y, \theta_k(\lambda) \mid \lambda). \quad (33)$$

We suppose that $\theta_k^*(\lambda)$ is not Pareto optimal. By the definition of Pareto optimality (Definition 2.2), there exists another personalized model $\theta_k'(\lambda) \in \Phi$ such that $\mathcal{L}(x, y, \theta_k'(\lambda)) < \mathcal{L}(x, y, \theta_k^*(\lambda))$. Therefore, we have the following inequality:

$$\mathcal{L}_i(x, y, \theta_k'(\lambda)) \leq \mathcal{L}_i(x, y, \theta_k^*(\lambda)), \forall i \in \{1, 2\}. \quad (34)$$

We also can obtain inequality (31), which contradicts the assumption, so θ_k^* is Pareto optimal for problem (10). $\square$

A.3.6 Proof of Proposition 5.7.

PROOF. We now prove an upper bound of the difference between the personalized model on any client (take client k as an example) ϕ_k and the optimal personalized model ϕ_k^* . Let I_k^t indicate if user k is selected at the t -round, and $\mathbb{E}[I_k^t] = p_k$. We have

$$\begin{aligned} & \mathbb{E}[\|\phi_k^{t+1} - \phi_k^*\|^2] \\ &= \mathbb{E}[\|\phi_k^t - \eta I_k^t \nabla \tilde{g}(\phi_k^t; \psi^t) - \phi_k^*\|^2] \end{aligned} \quad (35)$$

$$\begin{aligned} &= \mathbb{E}[\|\phi_k^t - \phi_k^*\|^2] + \eta^2 \mathbb{E}[\|I_k^t \nabla \tilde{g}(\phi_k^t; \psi^t)\|^2] \\ &\quad + 2\eta \mathbb{E}[p_k \nabla \tilde{g}(\phi_k^t; \psi^t)^T (\phi_k^* - \phi_k^t)] \end{aligned} \quad (36)$$

$$\begin{aligned} &\stackrel{(a)}{\leq} (1 - \eta p_k \check{\mu}) \mathbb{E}[\|\phi_k^t - \phi_k^*\|^2] + \underbrace{\eta^2 \mathbb{E}[\|\nabla \tilde{g}(\phi_k^t; \psi^t)\|^2]}_{C_1} \\ &\quad + \underbrace{2\eta p_k \mathbb{E}[\tilde{g}(\phi_k^*; \psi^t) - \tilde{g}(\phi_k^t; \psi^t)]}_{C_2}, \end{aligned} \quad (37)$$

where (a) is due to the smoothness of $\tilde{g}$. We then find the upper bounds of C_1 and C_2 , respectively. For C_1 ,

$$\begin{aligned} &\eta^2 \mathbb{E}[\|\nabla \tilde{g}(\phi_k^t; \psi^t)\|^2] \\ &= \eta^2 \mathbb{E}[\|\nabla \tilde{g}(\phi_k^t; \psi^t) - \nabla \tilde{g}(\phi_k^t; \psi^*) + \nabla \tilde{g}(\phi_k^t; \psi^*)\|^2] \\ &= 2\eta^2 \mathbb{E}[\|\nabla \tilde{g}(\phi_k^t; \psi^*)\| \|\nabla \tilde{g}(\phi_k^t; \psi^t) - \nabla \tilde{g}(\phi_k^t; \psi^*)\|] \\ &\quad + \eta^2 \mathbb{E}[\|\nabla \tilde{g}(\phi_k^t; \psi^*)\|^2] + \eta^2 \mathbb{E}[\|\nabla \tilde{g}(\phi_k^t; \psi^t) - \nabla \tilde{g}(\phi_k^t; \psi^*)\|^2] \end{aligned} \quad (38)$$

$$\begin{aligned} &\stackrel{(b)}{\leq} \eta^2 \mathbb{E}[\|\nabla \tilde{g}(\phi_k^t; \psi^*)\|^2] + 2\eta^2 \check{L} \mathbb{E}[\|\nabla \tilde{g}(\phi_k^t; \psi^*)\| \|\psi^t - \psi^*\|] \\ &\quad + \eta^2 \check{L}^2 \mathbb{E}[\|\psi^t - \psi^*\|^2], \end{aligned} \quad (39)$$

where (b) is due to the convexity of $\tilde{g}$. Then, we continue to find the upper bound of C_2 ,

$$\begin{aligned} &2\eta p_k \mathbb{E}[\tilde{g}(\phi_k^*; \psi^t) - \tilde{g}(\phi_k^t; \psi^t)] \\ &\stackrel{(c)}{\leq} 2\eta p_k \mathbb{E}[\tilde{g}(\phi_k^*; \psi^t) - \tilde{g}(\phi_k^t; \psi^t) + \eta p_k (\check{L} - \check{\mu}) \|\psi^t - \psi^*\|^2] \\ &\quad + \mathbb{E}[(\nabla \tilde{g}(\phi_k^*; \psi^t) - \nabla \tilde{g}(\phi_k^t; \psi^t))^T (\psi^t - \psi^*)] \end{aligned} \quad (41)$$

$$\begin{aligned} &\stackrel{(d)}{\leq} 2\eta p_k \mathbb{E}[\tilde{g}(\phi_k^*; \psi^t) - \tilde{g}(\phi_k^t; \psi^t) + \eta p_k (\check{L} - \check{\mu}) \|\psi^t - \psi^*\|^2] \\ &\quad + 2\eta p_k \check{L} \mathbb{E}[\|\phi_k^t - \phi_k^*\| \|\psi^t - \psi^*\|], \end{aligned} \quad (42)$$

where (c) uses the smoothness and convexity of $\tilde{g}$. (d) uses the convexity of $\tilde{g}$. So far, we have found the upper bounds of C_1 and C_2 . We then plugging their upper bounds Eq. (40) and Eq. (42) into Eq. (37), as follows:

$$\begin{aligned} &\mathbb{E}[\|\phi_k^{t+1} - \phi_k^*\|^2] \\ &\leq (1 - \eta p_k \check{\mu}) \mathbb{E}[\|\phi_k^t - \phi_k^*\|^2] + \eta^2 \mathbb{E}[\|\nabla \tilde{g}(\phi_k^t; \psi^*)\|^2] \\ &\quad + 2\eta^2 \check{L} \mathbb{E}[\|\nabla \tilde{g}(\phi_k^t; \psi^*)\| \|\psi^t - \psi^*\|] \\ &\quad + \eta^2 \check{L}^2 \mathbb{E}[\|\psi^t - \psi^*\|^2] + 2\eta p_k \mathbb{E}[\tilde{g}(\phi_k^*; \psi^t) - \tilde{g}(\phi_k^t; \psi^t)] \\ &\quad + 2\eta p_k \check{L} \mathbb{E}[\|\phi_k^t - \phi_k^*\| \|\psi^t - \psi^*\|] \\ &\quad + \eta p_k (\check{L} - \check{\mu}) \mathbb{E}[\|\psi^t - \psi^*\|^2] \end{aligned} \quad (43)$$

$$\begin{aligned} &\stackrel{(e)}{\leq} (1 - \eta p_k \check{\mu}) \mathbb{E}[\|\phi_k^t - \phi_k^*\|^2] + \eta^2 \mathbb{E}[\|\nabla \tilde{g}(\phi_k^t; \psi^*)\|^2] \\ &\quad + 2\eta^2 \check{L} \sqrt{\mathbb{E}[\|\nabla \tilde{g}(\phi_k^t; \psi^*)\|^2] \mathbb{E}[\|\psi^t - \psi^*\|^2]} \\ &\quad + \eta^2 \check{L}^2 \mathbb{E}[\|\psi^t - \psi^*\|^2] + 2\eta p_k \check{L} \sqrt{\mathbb{E}[\|\phi_k^t - \phi_k^*\|^2] \mathbb{E}[\|\psi^t - \psi^*\|^2]} \\ &\quad + \eta p_k (\check{L} - \check{\mu}) \mathbb{E}[\|\psi^t - \psi^*\|^2] \end{aligned} \quad (44)$$

$$\begin{aligned} &\stackrel{(f)}{\leq} (1 - \eta p_k \check{\mu}) \mathbb{E}[\|\phi_k^t - \phi_k^*\|^2] + \eta^2 G^2 + 2\eta^2 G \check{L} \sqrt{\mathbb{E}[\|\psi^t - \psi^*\|^2]} \\ &\quad + (\eta^2 \check{L}^2 + \eta p_k (\check{L} - \check{\mu})) \mathbb{E}[\|\psi^t - \psi^*\|^2] \\ &\quad + 2\eta p_k \check{L} \sqrt{\mathbb{E}[\|\phi_k^t - \phi_k^*\|^2] \mathbb{E}[\|\psi^t - \psi^*\|^2]}, \end{aligned} \quad (45)$$

where (e) is due to the Cauchy-Schwarz inequality and $\mathbb{E}[\tilde{g}(\phi_k^*; \psi^t) - \tilde{g}(\phi_k^t; \psi^t)] \leq 0$. (f) holds because the norm of the squared gradient is bounded by G^2 (Assumption 5.1). $\square$

A.3.7 Proof of Theorem 5.8.

PROOF. There exists $C < \infty$ such that for any client $k \in [K]$, $C > \frac{\mathbb{E}[\|\phi_k^0 - \phi_k^*\|^2]}{z(0)}$, we have

$$\mathbb{E}[\|\phi_k^0 - \phi_k^*\|^2] \leq Cz(0). \quad (46)$$

If $\mathbb{E}[\|\phi_k^t - \phi_k^*\|^2] \leq Cz(t)$ holds, then for $t + 1$,

$$\begin{aligned} &\mathbb{E}[\|\phi_k^{t+1} - \phi_k^*\|^2] \\ &\leq (1 - \frac{2z(t)}{A}) Cz(t) + \frac{z(t)^2}{A} \frac{4\sqrt{C}}{\check{\mu}} + \frac{z(t)^2}{A} \frac{2(\check{L} - \check{\mu})}{\check{\mu}} \\ &\quad + \frac{z(t)^2}{A} \frac{4}{Ap_k^2 \check{\mu}^2} (G^2 + \check{L}^2 z(t) + 2G\check{L}\sqrt{z(t)}) \end{aligned} \quad (47)$$

$$\stackrel{(a)}{\leq} (1 - \frac{2z(t)}{A}) Cz(t) + \frac{Cz(t)^2}{A} \quad (48)$$

$$= (1 - \frac{z(t)}{A}) Cz(t) \quad (49)$$

$$\stackrel{(b)}{\leq} Cz(t+1) \quad (50)$$

$$\stackrel{(c)}{\leq} C \cdot O(\log(\frac{1}{t+1})), \quad (51)$$

where (a) holds because C is a sufficiently large number such that coefficient of $\frac{z(t)^2}{A}$ is smaller than C . (b) is due to the condition $\frac{z(t+1)}{z(t)} \geq 1 - \frac{z(t)}{A}$. (c) is due to the Lemma 5.6. $\square$